

Pneumatic conveying characteristics in the rice husk powder industry and optimization of engineering process

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Highlights

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Conveying of [rice husk](#) powder is achieved by the positive pressure method.

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The mass flow rate achieved was 10 t/h, with a solid-gas ratio of 150 kg/kg.

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The conveying performance of rice husk powder has been studied and optimized.

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A design basis for the conveying of rice husk powder has been established.

Abstract

Large-scale biomass pneumatic conveying technology serves the fields of biomass reuse, including agricultural and forestry waste management, and [biomass gasification](#). This study focuses on the dense phase conveying characteristics of [rice husk](#) powder in a positive pressure [pneumatic conveying system](#), examining the impact of operational parameters on [mass flow rate](#) and solid-gas ratio. The study successfully achieved industrial dense-phase pneumatic conveying technology with a [mass flow rate](#) of 3000–10,000 kg/h and a solid-gas ratio as high as 160 kg/kg. A prediction [equation for mass flow rate](#) based on pressure drive was provided, which was suitable for this system and provided a design basis for pressure. Based on the design of [fluidization velocity](#), the design basis for the [fluidization](#) gas in actual operation was proposed. The phase diagram was used to analyze the conveying state and determine the economic [gas velocity](#). Furthermore, the economic [gas velocity](#) was predicted by using the function relationship

of *Ar* and *Re* numbers, and the prediction deviation is less than 5 %. This work provided a valuable reference for the efficient conveying and processing of biomass particles on a large scale.

Introduction

As a primary renewable energy source, biomass energy was widely utilized for generating heat energy, electricity, liquid fuel, and hydrogen. Biomass was abundant in nature and widely distributed, which had the advantages of renewable energy and low greenhouse gas emissions (Burton and Wu, 2016; Hensgen et al., 2014; Li et al., 2017); Popp et al. (2021); (Pradhan et al., 2018). Biomass energy could be harnessed through various methods such as direct combustion, gasification, and pyrolysis. Among these, biomass gasification technology has gained significant attention as a key approach for biomass utilization (Pang, 2019; Saleem, 2022; Sansaniwal et al., 2017; Shi et al., 2023; Yan et al., 2018a; Yan et al., 2018b). However, when attempting to introduce biomass particles into the reactor, several critical challenges arise, jeopardizing the continuous and stable operation of the entire system. Arch, perforation, and blockage were the most common issues encountered during biomass particle supply. Pneumatic conveying was a particle transportation technology utilizing airflow, which offered substantial technical advantages including high efficiency, environmental friendliness, and low energy consumption. This technology strongly supported the utilization and widespread adoption of biomass energy. Pneumatic conveying technology plays a pivotal role in the efficient and eco-friendly treatment of biomass and the removal of waste generated from agricultural and forestry operations. This technology enabled the transportation of biomass particles through a network of pneumatic conveying systems, which significantly enhanced the effectiveness and minimize energy consumption. Specifically, dense-phase pneumatic conveying exhibited remarkable efficiency, low gas conveying, and power consumption, making it highly competitive (Behera et al., 2023, Klinzing, 2018, Yang et al., 2011a, Zhou and Xiong, 2020a).

The demand for biomass energy utilization and the limitations of traditional conveying methods have led to the development of pneumatic conveying of biomass particles (Berruti and Briens, 2013). Mechanical feeders and pneumatic feeders were the two primary traditional transportation methods for biomass particles. The screw feeder was a representative equipment for mechanical feeders, which boasted high reliability and durability. However, the wear and tear of moving parts and the extrusion of biomass particles were major drawbacks. Additionally, the conveying capacity was restricted by the rotating speed of the screw machine, making it challenging to implement large-scale applications of biomass particles (Dai et al., 2012; Ma et al., 2024; Rahmanian- Koushkaki et al., 2020; Suri and Horio, 2009). In contrast, pneumatic feeders exhibited a simple

structure, flexible layout, simple operation, and easy maintenance(Wypych, 1999). However, biomass particles were prone to blockage at the silo outlet and pipeline interface, leading to unstable and discontinuous operation processes. Since the pneumatic feeder mainly belonged to the dilute phase, pipeline wear was significant, and pressure fluctuation was substantial(Berruti and Briens, 2013). When dealing with biomass particles, traditional mechanical conveying methods might encounter problems such as blockage, wear, significant fluctuations, and high energy consumption. Positive pressure pneumatic conveying regulated airflow on biomass particles, allowing for high-speed and continuous conveying, improving production efficiency, and reducing energy consumption and maintenance costs.

In the past, extensive research has been conducted on the pneumatic conveying characteristics of pulverized coal. The well-established pneumatic conveying technology for pulverized coal could serve as a valuable reference for the pneumatic conveying of biomass particles. Understanding the fundamental principles of pneumatic conveying was crucial as it guided industrial applications and practical operations. Zhou and Xiong (2020b) reported on the fundamental laws governing the dense-phase transportation of high-pressure pulverized coal. The study investigated the effects of gas volume on parameters such as mass flow rate and solid-gas ratio. Additionally, the pressure drop characteristic in the pipeline was an important parameter in the conveying process. Guo et al. (2013) examined the pressure drop characteristics of pulverized coal conveying in a horizontal pipe, while Zhou et al. (2011) studied the resistance characteristics of pulverized coal in a curved pipe. Jin et al. (2019) focused on the flow characteristics of pulverized coal in vertical pipes. Various measurement techniques have also been employed in the pneumatic conveying process. For instance, electrical capacitance tomography (ECT) technology could identify flow patterns in the pipeline(Cong et al., 2012b; Yang et al., 2011b). Electrostatic detection could determine the flow state in the pipeline(Gao et al., 2019), Real-time monitoring of the flow state in the pipeline could be achieved through acoustic measurement(He et al., 2016). The mature technologies and methods developed for the pneumatic conveying of pulverized coal serve as valuable references for the pneumatic conveying of biomass particles. Different from pulverized coal commonly used in gasification, biomass particles generally have the characteristics of irregular shape, large porosity and larger particle size. These characteristics can easily lead to particle deposition and blockage in the pipeline. These reasons limit the application of dense-phase pneumatic conveying of biomass particles.

The dense-phase pneumatic conveying technology of pulverized coal as gasification raw material was mature. However, the pneumatic conveying characteristics of biomass particles differ from those of pulverized coal. From the perspective of packing structure,

biomass particles have a low bulk density, resulting in a low solid-gas ratio and high energy consumption. Morphologically, biomass particles are irregular in shape, leading to poor flowability, which makes discharging from the silo challenging and represents one of the difficulties in dense-phase conveying. In terms of conveying methods, the variety of biomass types means there is currently no unified device or method capable of achieving dense-phase conveying of different biomass particles. Additionally, from the standpoint of gasification raw materials, biomass particles generally have a larger particle size, which can lead to deposition within the pipeline and increased risk of blockages. These factors highlight the challenges associated with the pneumatic conveying of biomass particles, underscoring the need to develop efficient dense-phase pneumatic conveying technologies specifically for biomass particles. The positive pressure pneumatic conveying technology has been developed in the field of pneumatic conveying of biomass particles. He et al. (2012) successfully conducted pneumatic conveying experiments with rice husk powder on a self-built platform, utilizing a system pressure above 2 MPa. However, the disadvantage was that the rice husk powder had a grain size of less than 100 microns and a solid-gas ratio of less than 20 kg/kg, indicating a dilute phase transportation process. Liang et al. (2015) achieved a mass flow rate of 1000 kg/h for 260 μm biomass particles at a system pressure above 3 MPa. They also compared the pneumatic conveying characteristics of biomass particles with pulverized coal particles in a high-pressure system. It was observed that biomass particles exhibited higher conveying start-up speed and greater resistance in the pipeline compared to pulverized coal particles. The pneumatic conveying characteristics of sawdust particles in horizontal pipes were investigated by Huang and Jin (2020). They discovered that the solid friction coefficient decreased with an increase in the dimensionless number and established an empirical relationship for pressure drop. Currently, biomass pneumatic conveying technology faces several challenges. The particles used for transportation were typically less than 1 mm in size. The high energy consumption required for crushing biomass particles posed a significant challenge in biomass gasification. Therefore, developing pneumatic conveying processes for larger particles can help reduce the energy consumption associated with biomass particle utilization (Kruszelnicka et al., 2020; Mani et al., 2004). Hydraulic transportation of biomass particles was a common method, but it also had the problem of high energy consumption (Faghani et al., 2020; Javed et al., 2022; Javed et al., 2021). Moreover, most biomass pneumatic transportation operates at the dilute phase level, resulting in high energy consumption. The mass flow rate of biomass pneumatic conveying was generally low, often below 1 t/h. These crucial issues ultimately limited the large-scale gasification application of biomass particles (Estejab et al., 2017; Kamaruzaman et al., 2023; Massaro Sousa et al., 2021).

As gasification particles, biomass particles have a larger particle size than traditional pulverized coal particles. This size discrepancy introduces numerous challenges in the dense-phase pneumatic conveying of such biomass particles. Consequently, developing high-efficiency dense-phase pneumatic conveying technology for these biomass particles is a pressing concern and a critical link that requires immediate resolution within the context of biomass gasification processes. In response to these challenges, the objectives of this study were outlined as follows: (1) To construct an experimental platform tailored for the dense phase conveying of biomass particles. The experiment utilized rice husk powder as the raw material to ascertain the viability of high-flow rate dense-phase pneumatic conveyance for biomass particles. (2) To investigate the conveying characteristics of rice husk powder and to analyze the impact of operational conditions on the conveyance performance. (3) To establish a design basis for three-way gas conveyance, utilizing the data of rice husk powder as a case study, and to propose a more comprehensive design framework that can accommodate a wider range of pneumatic conveying scenarios.

Experimental materials

As a commonly used biomass powder, rice husk powder was frequently employed in the study of biomass properties. Rice husk powder is widely used in composite materials, construction, environmental protection and biomass gasification (Bera and Kale, 2006; Feng et al., 2023; Li et al., 2022; Maksum et al., 2017; Su and Xu, 2024; Yun, 2022). The experimental material used in this study was rice husk powder purchased from the market. A multifunctional particle size analyzer (Camsizer XT, Retsch

Influence of pressurized gas on conveying characteristics

Pressurized gas plays an important role in maintaining the pressure of the feeding tank and providing stable power for the transportation process. Additionally, some of the gas would penetrate the powder bed and carry the powder into the pipeline. Therefore, it was crucial to study the impact of pressurized gas on the tank pressure. In the experiment, the fluidized gas was kept at a constant rate of 5 Nm³/h and the supplemented gas at 7 Nm³/h. The pressurized gas was then adjusted to a range of

Conclusions

This paper investigated the dense-phase conveying technology of rice husk powder in a positive pressure pneumatic conveying system to achieve efficient pneumatic conveying of biomass particles. The industrial design basis of pneumatic conveying based on the basic physical properties of particulate matter is innovatively realized in this paper. In comparison to previous studies on biomass pneumatic conveying, the following conclusions can be drawn.

- (1)

By implementing differential pressure

CRediT authorship contribution statement

Lu Haifeng: Writing – review & editing, Resources, Project administration, Methodology, Investigation. **Guo Xiaolei:** Funding acquisition, Formal analysis, Data curation, Conceptualization. **Liu Haifeng:** Validation, Supervision, Resources, Project administration, Methodology. **Du Hui:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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